BEU CONNECT

Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Time: 03 Hours

Code: 104602 ***Subject: Digital Communication*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

(a) The sampling theorem is essential to avoid which problem?
- (i) Crosstalk
- (ii) Quantization error
- (iii) Attenuation
- (iv) Aliasing

(b) In Time Division Multiplexing (TDM), multiple signals share:
- (i) Power slots
- (ii) Code slots
- (iii) Time slots
- (iv) Frequency slots

(c) Which modulation technique is most commonly used in CDMA?
- (i) QAM
- (ii) PSK
- (iii) ASK
- (iv) FSK

(d) In Quadrature Amplitude Modulation (QAM), the information is carried by:
- (i) Frequency and phase
- (ii) Amplitude and phase
- (iii) Amplitude only
- (iv) Phase only

(e) What does the signal-to-quantization noise ratio (SQNR) measure?
- (i) Bandwidth efficiency
- (ii) Power efficiency
- (iii) Distortion in quantized signals
- (iv) Signal attenuation

(f) Which line coding technique uses zero crossing for binary '1' and '0'?
- (i) AMI
- (ii) Manchester coding
- (iii) RZ
- (iv) NRZ

(g) Which of the following is true for Adaptive Delta Modulation (ADM)?
- (i) It uses fixed bandwidth
- (ii) It increases quantization noise
- (iii) It reduces slope overload
- (iv) It has fixed step size

(h) A matched filter maximizes which of the following?
- (i) Transmission speed
- (ii) Signal-to-noise ratio
- (iii) Noise
- (iv) Signal power

(i) Inter-symbol interference (ISI) primarily occurs due to:
- (i) High bit rate
- (ii) Low signal-to-noise ratio
- (iii) Over-sampling
- (iv) Improper filtering

(j) Amplitude Shift Keying (ASK) modulates which signal characteristic?
- (i) Pulse width
- (ii) Frequency
- (iii) Phase
- (iv) Amplitude

Q.2 Solve both questions : **[7+7=14]**

(a) State and prove the sampling theorem. Why is it important in digital communication systems?

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(b) Explain the working principle of Differential Phase Shift Keying (DPSK) and how it differs from standard PSK.

Q.3 Solve both questions : **[7+7=14]**

(a) A signal is encoded using 8-bit PCM. Calculate the data rate if the signal is sampled at 8 kHz. What will be the data rate for 16-bit PCM at the same sampling rate?

(b) Derive the expression for the probability of error in a matched filter receiver.

Q.4 Solve both questions : **[7+7=14]**

(a) Discuss the concept of a baseband signal receiver and the factors affecting the probability of error in digital communication systems.

(b) Explain the process of quantization in a digital communication system. Differentiate between uniform and non-uniform quantization.

Q.5 Solve both questions : **[7+7=14]**

(a) A QPSK system transmits at a data rate of 2 Mbps. If the noise power spectral density is 0.5μW/Hz and the signal power is 1 mW, calculate the bit error rate (BER) of the system.

(b) Discuss the principles of Code Division Multiple Access (CDMA) and how it differs from TDM and FDM.

Q.6 Solve both questions : **[7+7=14]**

(a) Derive the Nyquist criterion for zero ISI. Explain how this criterion ensures distortion-free transmission.

(b) Describe the block diagram of a digital communication system and function of each block.

Q.7 Solve both questions : **[7+7=14]**

(a) What is aliasing? Explain the causes of aliasing and how it can be avoided in digital communication systems.

(b) Discuss the various types of line coding techniques used in digital communication. Compare unipolar, polar, and bipolar schemes.

Q.8 Solve both questions : **[7+7=14]**

(a) Explain Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) with waveform examples.

(b) What is Minimum Shift Keying (MSK)? Explain its advantages over FSK and other digital modulation techniques.

Q.9 Write short notes on any two of the following: **[14]**

(a) (a) Orthogonal Frequency Division Multiple Access (OFDMA).

(b) (b) Matched filter.

(c) (c) Coherent reception & Non-Coherent reception.

(d) (d) Advantages of Digital communication system over Analog communication systems.

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